

Visual Prototype Conditioned Focal Region Generation for UAV-Based Object Detection

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Abstract

Unmanned aerial vehicle (UAV) based object detection is a critical but challenging task, when applied in dynamically changing scenarios with limited annotated training data. Layout-to-image generation approaches have proved effective in promoting detection accuracy by synthesizing labeled images based on diffusion models. However, they suffer from frequently producing artifacts, especially near layout boundaries of tiny objects, thus substantially limiting their performance. To address these issues, we propose UAVGen, a novel layout-to-image generation framework tailored for UAV-based object detection. Specifically, UAVGen designs a Visual Prototype Conditioned Diffusion Model (VPC-DM) that constructs representative instances for each class and integrates them into latent embeddings for high-fidelity object generation. Moreover, a Focal Region Enhanced Data Pipeline (FRE-DP) is introduced to emphasize object-concentrated foreground regions in synthesis, combined with a label refinement to correct missing, extra and misaligned generations. Extensive experimental results demonstrate that our method significantly outperforms state-of-the-art approaches, and consistently promotes accuracy when integrated with distinct detectors. The source code is available at <https://github.com/Sirius-Li/UAVGen>.

1. Introduction

Unmanned Aerial Vehicle (UAV) based object detection [7, 21, 28, 30, 36, 52], which localizes and identifies targets from aerial perspectives, has demonstrated practical value across a range of applications [12, 19, 20, 41]. However, as deep learning has become the mainstream paradigm, UAV-based detection remains fundamentally constrained by the

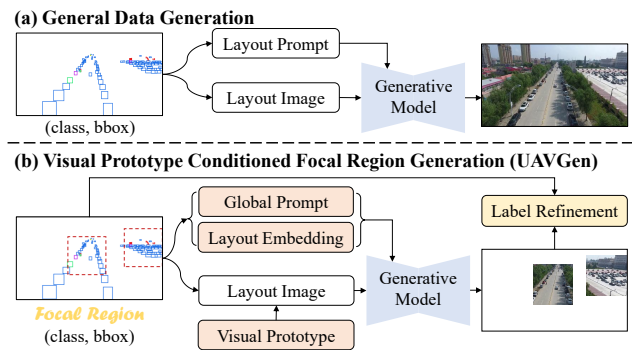


Figure 1. Illustration of different layout-to-images data generation methods. (a) In general data generation, the layout information is directly derived from real data, and synthesized data is directly used as training data for the detector. (b) Our UAVGen enhances this paradigm by introducing multi-cue, high-quality layout conditions for focal regions, as well as refining the synthesized data.

scarcity of high-quality training data, especially in dynamically changing environments [6, 8, 9]. Consequently, its robustness and generalization capabilities still fall short of those achieved in general object detection tasks.

In order to obtain sufficient and diverse training data under limited cost, data augmentation has emerged as a promising solution [11, 14, 53]. Early approaches construct additional samples by applying geometric transformations to existing images, such as flipping, rotation or simple cropping and paste [15, 59]. Recently, benefiting from the generalization capabilities of generative models, diffusion-driven data augmentation has emerged as a new paradigm that synthesizes images from input layouts (referred to as layout-to-image), producing entirely novel variations in appearance. By textual prompt conditioning [3, 51], latent-space injection [46], or layout-image conditioning [62], it has proved effective in enhancing detector training under limited-data scenarios. Nonetheless, despite their success on general object detection benchmarks, such approaches have not yet

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achieved comparable gains in UAV-based detection.

We attribute this performance gap to three key limitations inherent to UAV image characteristics: 1) *Low-quality visual layouts with small and overlapped targets*. Limited by flight altitude and fixed viewing geometry, UAV-captured objects often present small scales and frequent overlaps. Such regions tend to produce blurred or visually entangled layout representations, which impair the clarity of fine-grained conditional signals, interfere with diffusion model training, and lower the fidelity of synthesized samples. 2) *Inefficient utilization of model capacity under highly uneven spatial distribution*. Objects in UAV imagery are typically concentrated in a small fraction of the scene, leaving large regions with negligible informative. Diffusion models inadvertently allocate excessive capacity to these low-information regions, weakening their ability to capture fine-grained object characteristics. 3) *Inconsistency between synthesized images and ground-truth annotations*. Due to the inherent stochasticity of the diffusion process, existing methods inevitably generate images deviate from the input layouts, resulting in missing generations, extra generations and label misalignments. Such inconsistencies are further amplified in small-object-dominated UAV scenarios, introducing label noise that impairs the training of detectors.

To address these challenges, we propose UAVGen, a novel diffusion-driven data augmentation framework for UAV-based object detection. To improve layout condition quality, we first design a visual prototype conditioned diffusion model (VPC-DM). Specifically, a dual-criteria selection mechanism extracts visually clear, semantically well-defined objects from annotated data to construct high-fidelity visual prototypes. With these prototypes, we build layout images integrated with global and fine-grained textual layout semantics, delivering more robust guidance for image generation. Second, we develop a focal region enhanced data pipeline (FRE-DP). Instead of conventional full-image generation, it focuses on densely clustered small-object regions, steering both diffusion model and detector optimization toward detection-critical areas. Finally, a label refinement step corrects annotations of synthesized images, providing more precise training data for detectors.

Our main contributions are summarized in three-fold:

- We propose a novel framework to synthesize data for UAV-based object detection, dubbed UAVGen. To the best of our knowledge, this work is the first data synthesis approach for UAV-based detector training.
- We design a visual prototype conditioned diffusion model as well as a focal region enhanced data pipeline, to address the problems of low layout-quality, misfocused optimization and image-label inconsistency, thereby facilitating the UAV-based detector training.
- Comprehensive experiments on multiple benchmarks demonstrate that our method consistently improves object

detection performance, achieving state-of-the-art results across both general and UAV-based detectors.

2. Related Work

2.1. UAV-based Object Detection

Object detection in UAV aerial imagery faces severe challenges including tiny objects, complex backgrounds, and strict computational constraints. Although general-purpose detectors such as the YOLO series [24, 35, 36, 47] and GFL [30] provide efficient foundational frameworks, they often underperform in aerial contexts without domain-specific modifications. To mitigate these issues, various specialized detectors have been developed. CEASC [7] employs global context-enhanced normalization and adaptive sparse convolution for efficient small object detection. UFPMP-Det [21] utilizes foreground packing and multi-proxy networks to jointly optimize accuracy and speed. RemDet [28] adopts residual context mining to enhance the representation of tiny-objects features and reduce the decay of deep-layer information.

2.2. Diffusion Models

Diffusion models have emerged as a leading paradigm in generative modeling due to their stability and superior image synthesis capabilities [5, 18, 26, 34, 40]. Unlike earlier frameworks such as Generative Adversarial Networks (GANs) [2, 16, 23] and Variational Autoencoders (VAEs) [27, 38, 49], diffusion models generate data through a progressive denoising process. They iteratively refine samples from Gaussian noise to approximate the data distribution [43–45]. Recent efforts focus on improving both efficiency and controllability. Latent Diffusion Models (LDMs) [40] perform denoising in latent space, boosting computational efficiency while preserving image fidelity. Extensions like ControlNet [57] enable structural conditioning using segmentation masks and layout annotations, allowing finer control over outputs.

2.3. Layout-to-Image Generation

Generating photorealistic images from abstract layouts has attracted increasing attention. Early layout-to-image (L2I) methods based on GANs [32] or transformers [22], conditioned on bounding boxes and labels, often suffered from instability and limited realism in complex scenes [32]. Diffusion-based approaches alleviate these issues by enhancing both fidelity and spatial alignment. LayoutDiffusion [60] and LayoutDiffuse [4] incorporate layout-aware attention into LDMs to better model object interactions. Other frameworks, including MultiDiffusion [1], BoxDiff [54], GLIGEN [31], ReCo [55], and GeoDiffusion [3], further improve layout conditioning via training-free sampling or architectural enhancements. On a finer level, MIGC [61]

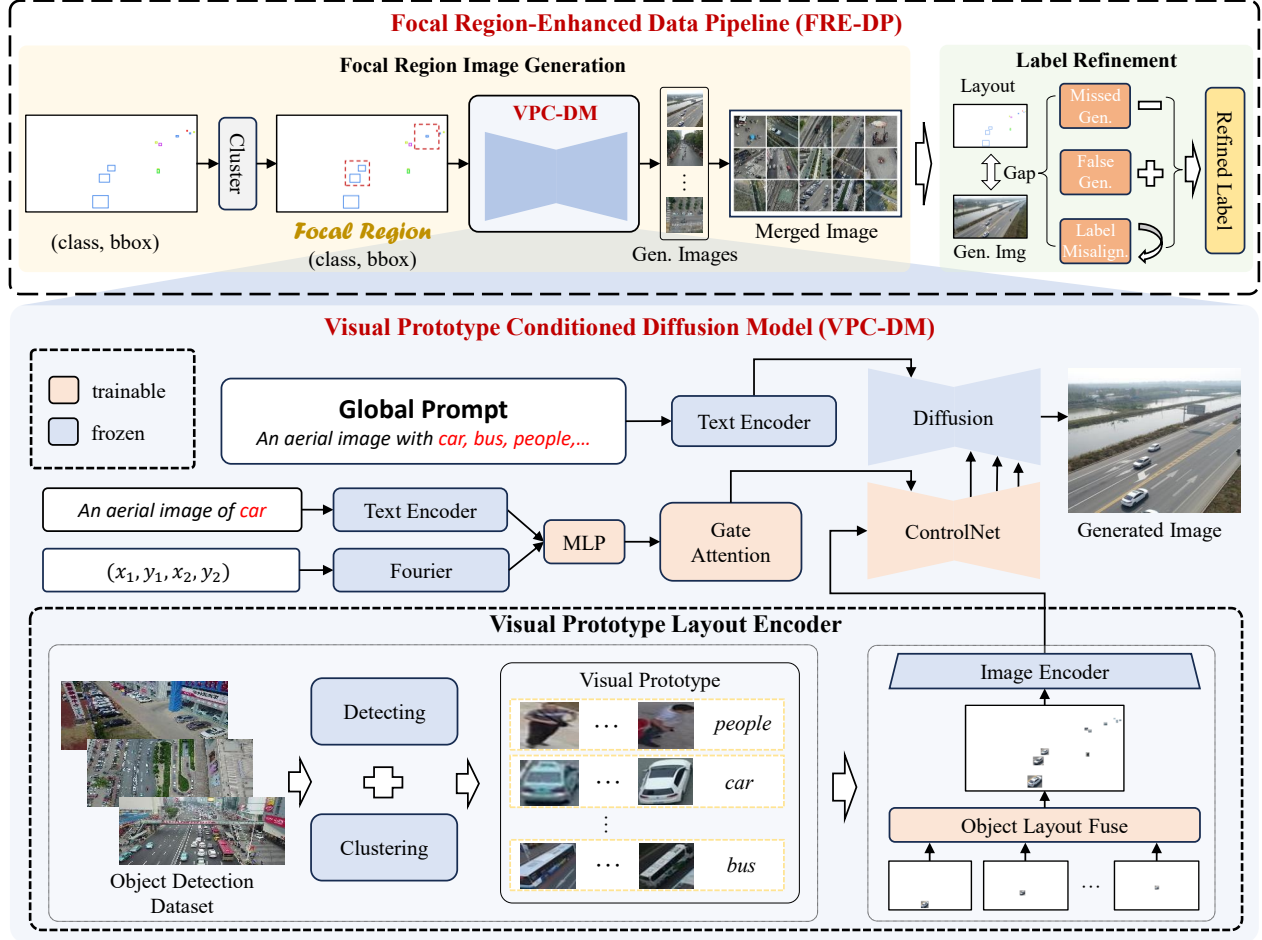


Figure 2. **Architecture of Visual Prototype Conditioned Focal Region Generation.** (a) Virtual Prototype Conditioned Diffusion Model (VPC-DM) generates images guided by layout images which is produced from selected visual prototypes. (b) Focal Region-Enhanced Data Pipeline (FRE-DP) synthesizes images on object-centric areas to avoid limitation of small object generation. Moreover, Label Refinement mitigates the misalignment between layouts and generated images.

and InstanceDiffusion [50] introduce instance-aware control using boxes, masks, or scribbles, enabling more diverse and spatially coherent synthesis.

2.4. Synthetic Data Generation

Copy-Paste methods [10, 11, 15, 59] synthesize training data by randomly placing foreground objects onto background images, but produce noticeable artifacts along object boundaries. Rendering-based methods [25, 39, 42, 56] synthesize images from 3D simulators to enhance detection, which suffer from domain gap between rendered and real UAV imagery. Recent efforts turn to diffusion-based models to mitigate these limitations. DatasetGAN [58], DatasetDM [53], and Satsynth [48] produce high-quality annotated data for object detection. However, challenges remain for small objects, image blurriness, and dense backgrounds—particularly in UAV imagery. Task-specific meth-

ods like TrackDiffusion [29], MagicDrive [14], and AeroGen [46] improve spatial consistency through multi-object tracking and 3D perception. Yet, occlusion, interference, and scale variation still pose obstacles in aerial scenes.

3. Method

3.1. Framework Overview

We follow the typical paradigm of diffusion-driven data augmentation for object detection, conducting a framework specifically tailored for UAV-based detection tasks. As for the overall process, let the real-world UAV-based detection dataset be denoted as $\mathcal{D}^{\text{real}} = \{(I_i^{\text{real}}, L_i^{\text{real}})\}_{i=1}^N$, consisting of N image-annotation pairs. Each image I_i^{real} is associated with an annotation set $L_i^{\text{real}} = \{(b_j^{\text{real}}, c_j^{\text{real}})\}_{j=1}^{n_i}$, which contains n_i bounding boxes b_j^{real} with their corresponding category labels $c_j^{\text{real}} \in \{1, 2, \dots, C\}$. A diffusion model $G_\theta(\cdot)$

with parameter θ is trained on $\mathcal{D}^{\text{real}}$ to synthesize a new images conditioned on the layout L_i^{real} and auxiliary information $\mathcal{C}_i$, thereby forming a synthetic dataset $\mathcal{D}^{\text{syn}}$ as Eq. (1).

$$I_i^{\text{syn}} = G_\theta(L_i^{\text{real}}, \mathcal{C}_i), \quad \mathcal{D}^{\text{syn}} = \{(I_i^{\text{syn}}, L_i^{\text{real}})\}_{i=1}^M. \quad (1)$$

Finally, $\mathcal{D}^{\text{syn}}$ and $\mathcal{D}^{\text{real}}$ are combined for detector training:

$$\phi^* = \arg \min_{\phi} \mathcal{L}_{\text{det}}(F_\phi(\mathcal{D}^{\text{real}} \cup \mathcal{D}^{\text{syn}})), \quad (2)$$

where $F_\phi(\cdot)$ denotes the desired object detector with trainable parameter ϕ and $\mathcal{L}^{\text{det}}$ is the standard detection loss.

However, such general paradigm encounters challenges in UAV scenarios due to the previously discussed complex imaging conditions, including small scale, uneven spatial distribution and overlapping. To address these issues, we introduce an enhanced data synthetic framework, namely UAVGen, which improves the fidelity and label consistency of $\mathcal{D}^{\text{syn}}$, thereby facilitating the optimization of Eq. (2). Specifically, we first introduce a Visual Prototype Conditioned Diffusion Model (VPC-DM) to produce more realistic UAV imagery. Subsequently, a Focal Region-Enhanced Data Pipeline (FRE-DP) is designed to further synthesize the generated images and improve their consistency with corresponding labels.

3.2. Visual Prototype Conditioned Diffusion Model

Dual-Criterion Visual Prototype Selection. When training $G_\theta(\cdot)$ for conditioned generation, a common practice is directly cropping all object regions L_i^{real} from $\mathcal{D}^{\text{real}}$ to serve as supervision. However, due to the small-scale appearance and dense distribution with overlap of UAV objects, these cropped patches are prone to suffer from noise and low quality, yielding unstable diffusion training. To ensure reliable supervision, we instead employ high-quality target regions, referred to as visual prototypes, which are selected sequentially according to two criteria.

First, we retain prototypes with clear appearance and accurate localization in the visual space. Taking advantage of the strong generalization capability, we use a pre-trained detector $D(\cdot)$ to collect all detected regions from $\mathcal{D}^{\text{real}}$, and then group them by predicted class. This results in $\mathcal{G}^c = \{(b_i^{\text{det}}, s_i)\}_{i=1}^{K_c}$ for each class c , where b_i^{det} denotes the detected bounding box and s_i is its confidence score. Noting that the confidence score for each class c typically follows a normal distribution $\mathcal{N}(\mu_c, \sigma_c^2)$, we select high-confidence and well-localized candidates according to:

$$\mathcal{V}^c = \{b_i^{\text{real}} | (b_i^{\text{det}}, s_i) \in \mathcal{G}^c, \text{IoU}(b_i^{\text{real}}, b_i^{\text{det}}) \geq \tau^{\text{det}}, s_i \geq \Phi_c^{-1}(\alpha)\}, \quad (3)$$

where τ^{det} is the IoU threshold, and $\Phi_c^{-1}(\alpha)$ represents the α -quantile of the confidence distribution for $\mathcal{G}^c$.

We further refine the candidates in the latent space to clarify fine-grained class boundaries and reduce generation confusion. Concretely as shown in Eq. (4), for each candidate region $b \in \mathcal{V}^c$, we extract the corresponding image patch and encode it using a VAE encoder $\mathcal{E}_{\text{img}}(\cdot)$ to obtain a latent representation. Then, the visual prototype set $\mathcal{P}^c$ for class c is constructed by selecting candidates whose latent embeddings lie close to the class-specific centroid μ^c .

$$\mathcal{P}^c = \{b \in \mathcal{V}^c \mid \|\mathcal{E}_{\text{img}}(b) - \mu^c\|^2 < \tau^{\text{lat}}\}, \quad (4)$$

$$\mu^c = \frac{1}{|\mathcal{V}^c|} \sum_{b \in \mathcal{V}^c} \mathcal{E}_{\text{img}}(b).$$

Multi-Source Condition Encoding. To achieve controllable and semantically consistent generation, we design a Multi-Source Condition Encoding mechanism that provides complementary conditioning from both visual-prototype-enhanced layouts and textual information.

Given a layout L_i^{real} as defined in Sec. 3.1, for each region $(b_j^{\text{real}}, c_j^{\text{real}}) \in L_i^{\text{real}}$, we sample a visual prototype P_j from the corresponding class-specific set $\mathcal{P}^c$, where $c = c_j^{\text{real}}$. Then, P_j is placed at position b_j^{real} with geometrical transformation to match the target scale, forming a composite image I_j^{blank} on a blank (*i.e.* zero-valued) canvas. Finally, we obtain a layout embedding v_i , which is enhanced by visual prototype, by aggregating the latent representations of all regions through a 3D convolutional network $\text{Conv}(\cdot)$ formally expressed as Eq. (5). The module fuses features along the prototype dimension: N features of shape $B \times C \times H \times W$ are stacked to $B \times N \times C \times H \times W$, fused to $B \times 1 \times C \times H \times W$. Learnable padding handles cases with fewer than N prototypes.

$$v_i = \text{Conv}(\mathcal{E}_{\text{img}}(I_1^{\text{blank}}), \mathcal{E}_{\text{img}}(I_2^{\text{blank}}), \dots, \mathcal{E}_{\text{img}}(I_{n_i}^{\text{blank}})). \quad (5)$$

Furthermore, to complement visual cues with class semantics, we construct layout-aware textual embeddings. Specifically, we introduce two textual prompts describing the overall scene and individual objects, respectively:

$$\begin{aligned} t_i^g &= \text{'An aerial image with } \{c_1\}, \{c_2\}, \dots, \{c_{n_i}\}. \text{' } \\ t^{c_j} &= \text{'An aerial image of } \{c_j\}. \text{' } \end{aligned} \quad (6)$$

As shown in Eq. (7), t_i^g is directly encoded by a text encoder $\mathcal{E}_{\text{text}}(\cdot)$ to gain the global embedding e_i^g . To capture fine-grained, object-level semantics, we enrich each object embedding with positional information via Fourier embeddings $\mathcal{F}(\cdot)$, inspired by GLIGEN [31]. Consequently, each object-specific text embedding is concatenated with its corresponding positional embedding and processed through an MLP-based network, after which all n_i features are aggregated by a gated attention network $GA(\cdot)$, resulting in a unified fine-grained layout embedding e_i^f .

$$\begin{aligned} e_i^g &= \mathcal{E}_{\text{text}}(t_i^g), \\ e_i^f &= GA\left(\left\{\text{MLP}\left(\left[\mathcal{E}_{\text{text}}(t^{c_j}); \mathcal{F}(b_j^{\text{real}})\right]\right)\right\}_{j=1}^{n_i}\right). \end{aligned} \quad (7)$$

Condition Injection and Training. Building upon the standard diffusion process parameterized by the noisy latent $\mathbf{x}_t$ and timestep t , we incorporate the previously obtained multi-source conditions to enhance controllable and layout-consistent generation. Specifically, the visual prototype-based layout embedding $\mathbf{v}_i$ and the fine-grained textual embedding $\mathbf{e}_i^f$ are injected through a ControlNet $CN(\cdot)$, which produces a layout-aware conditional feature $\mathcal{C}_i$ as:

$$\mathcal{C}_i = CN(\mathbf{x}_t, t \mid \mathbf{v}_i, \mathbf{e}_i^f). \quad (8)$$

Denoting ϵ_θ as the noise prediction network of G_θ , both $\mathbf{e}_i^g$ and $\mathcal{C}_i$ jointly guide the denoising process by ϵ_θ . This enables the generative process to maintain the overall scene semantics as well as preserving fine-grained object layouts.

To further enhance the fidelity of target objects, we employ a foreground-aware reweighted loss that emphasizes supervision on object regions:

$$\mathcal{L}_{\text{layout}} = \mathbf{w} \odot \mathbb{E}_{\mathbf{x}_{0,t}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t \mid \mathbf{e}_i^g, \mathcal{C}_i)\|^2], \quad (9)$$

where $\mathbf{w}$ is a spatial weighting map derived from L_i^{real} . Pixels within object regions are assigned a higher weight (*i.e.* > 1), while background regions retain a unit weight.

3.3. Focal Region Enhanced Data Pipeline.

Region-based Data Synthesis. UAV-based images typically exhibit small-object clustering, with the majority of image regions containing few or even no target instances. Such sparsely populated images may both limit the effectiveness of diffusion-based generation and reduce the training efficiency of object detection. To address this issue, we concentrate on object-rich areas, termed focal regions, to enhance both data synthesis and detector training.

Given a layout L_i^{real} as defined in Sec. 3.1, we first compute the geometric center of each b_j^{real} , denoted as $\mathbf{p}_j$. A K-means clustering is then applied to the set of centroids $\{\mathbf{p}_j\}_{j=1}^{n_i}$ to obtain K cluster centers $\{\mathbf{m}_k\}_{k=1}^K$ such that:

$$\mathbf{m}_k = \arg \min_{\mathbf{m}} \sum_{\mathbf{p}_j \in \mathcal{S}_k} \|\mathbf{p}_j - \mathbf{m}\|^2, \quad (10)$$

where $\mathcal{S}_k$ is the set of centroids assigned to cluster k .

Next, for each $\mathbf{m}_k$, the corresponding focal region $\mathcal{B}_k$ is determined by solving an overlap maximization problem:

$$\mathcal{B}_k = \arg \max_{\mathcal{B} \in \Omega(\mathbf{m}_k)} \sum_{j=1}^{n_i} \mathbb{I}[b_j^{\text{real}} \cap \mathcal{B} = b_j^{\text{real}}], \quad (11)$$

where $\Omega(\mathbf{m}_k)$ is the set of candidate regions with identical size containing $\mathbf{m}_k$.

The selected object-dense regions are then cropped to construct a high-information-density dataset $\mathcal{D}^{\text{real.dense}}$, which serves as a refined substitute for $\mathcal{D}^{\text{real}}$ and is subsequently used as the training and generation target for our proposed VPC-DM (*i.e.* Sec. 3.2).

Finally, the generated focal-region images are merged back into the original image resolution to form a high-information-density synthetic dataset $\mathcal{D}^{\text{syn.dense}}$ as an alternative of the typical $\mathcal{D}^{\text{syn}}$. This dataset is then used to train the detector as formulated in Eq. (2).

Label Refinement. Despite advances by our proposed layout-to-image synthesis framework, discrepancies between synthesized images and input layouts remain a significant challenge, especially for UAV imagery characterized by small-scale objects. We categorize these inconsistencies into three types: *missed generations*, *false generations*, and *label misalignments*. To mitigate these issues, we propose a post-processing pipeline with label refinement.

Given a synthesized sample $(I^{\text{syn}}, L^{\text{real}}) \in \mathcal{D}^{\text{syn.dense}}$ obtained above (for simplicity, the subscript ‘dense’ is omitted), we apply a pretrained detector $D(\cdot)$ on I^{syn} , obtaining predicted results:

$$L^{\text{det}} = D(I^{\text{syn}}) = \{(b_j^{\text{det}}, c_j^{\text{det}})\}_{j=1}^N. \quad (12)$$

We then perform IoU-based matching with threshold τ^{ref} between predicted boxes and real boxes, resulting in:

$$\mathcal{M} = \{(b_i^{\text{real}}, b_j^{\text{det}}) \mid \text{IoU}(b_i^{\text{real}}, b_j^{\text{det}}) \geq \tau^{\text{ref}}\}. \quad (13)$$

First, missed generations refer to regions where the diffusion model fails to generate the corresponding object, denoted as L^{miss} . To discard missing instances from labels and filter out low-quality ones, we preserve the labels aligned with detection results L^{det} and discard low-quality ones with threshold α on corresponding detection confidence scores. The refined label set is updated as Eq. (14).

$$\begin{aligned} L^{\text{miss}} &= L^{\text{real}} \setminus \{(b_i^{\text{real}}, c_i^{\text{real}}) \mid (b_i^{\text{real}}, b_j^{\text{det}}) \in \mathcal{M}\}, \\ L^{\text{ref}} &= \{(b_i^{\text{real}}, c_i^{\text{real}}) \in L^{\text{real}} \setminus L^{\text{miss}} \mid s_j \geq \Phi_{c_j}^{-1}(\alpha)\} \end{aligned} \quad (14)$$

Second, false generations represent extra generated objects, denoted as L^{false} . Thus, we apply a confidence filter per-class and update the refined set as Eq. (15).

$$\begin{aligned} L^{\text{false}} &= L^{\text{det}} \setminus \{(b_j^{\text{det}}, c_j) \mid (b_i^{\text{real}}, b_j^{\text{det}}) \in \mathcal{M}\}, \\ L^{\text{ref}} &\leftarrow L^{\text{ref}} \cup \{(b_j^{\text{det}}, c_j^{\text{det}}) \in L^{\text{false}} \mid s_j \geq \Phi_{c_j}^{-1}(\beta)\}. \end{aligned} \quad (15)$$

Third, for label misalignments, we focus on mitigating label discrepancies by refining each $(b_j^{\text{ref}}, c_j^{\text{ref}}) \in L^{\text{ref}}$ as Eq. (16) with γ controlling reliance on predictions.

$$(b_j^{\text{ref}}, c_j^{\text{ref}}) \leftarrow \begin{cases} (b_j^{\text{det}}, c_j^{\text{det}}) & \text{if } s_j > \Phi_{c_j}^{-1}(\gamma) \\ (b_j^{\text{ref}}, c_j^{\text{ref}}) & \text{otherwise} \end{cases}. \quad (16)$$

Ultimately, the refined dataset $\mathcal{D}^{\text{ref}} = (I^{\text{syn}}, L^{\text{ref}})$ is employed to construct the training set for object detector. α , β , and γ are set according to the accuracy of detector $D(\cdot)$.

Table 1. Comparison of FID scores and Average Precision (%) of UAV-based detector trained with augmented data produced by different methods on VisDrone and UAVDT. “Real only” means the detector is trained only on real data. The best results are highlighted in **bold**.

Dataset	Method	FID↓	Average Precision↑					
			mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
VisDrone [9]	Real only	-	24.5	42.1	24.5	15.4	36.1	37.9
	CopyPaste [11]	-	23.6	41.0	23.5	14.7	34.6	40.2
	GLIGEN [31]	72.68	24.8	43.0	24.6	15.6	36.3	39.0
	Geodiffusion [3]	57.96	24.7	42.9	24.7	15.4	36.1	39.1
	AeroGen [46]	48.04	24.9	43.3	24.5	15.6	36.3	43.3
	UAVGen (Ours)	34.34	25.9	44.8	26.0	16.7	37.5	43.7
UAVDT [8]	Real only	-	14.5	26.1	14.7	10.3	24.1	11.8
	Geodiffusion [3]	47.81	14.0	28.4	12.1	10.3	24.2	12.2
	AeroGen [46]	31.99	15.0	28.3	14.5	10.7	25.1	12.6
	UAVGen (Ours)	29.73	16.6	30.9	16.1	11.7	27.8	14.8

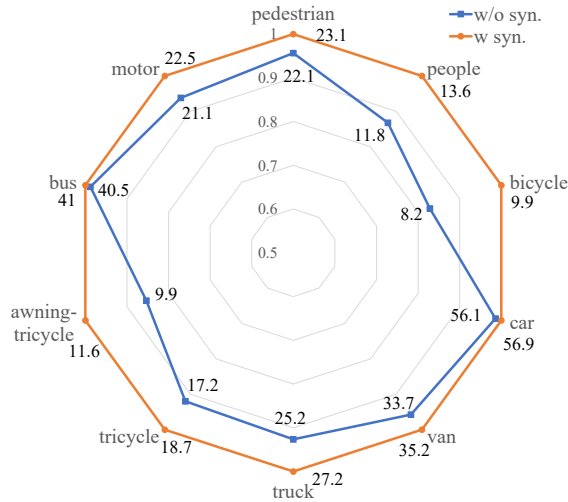


Figure 3. Comparison of mAP across different categories on VisDrone. Our method (w syn.) generates 738 synthesized images, yields consistent performance improvements across all categories when compared to the non-augmented baseline (w/o syn.).

4. Experimental Results and Analysis

4.1. Experimental Settings

Dataset and Evaluation Metrics. We evaluate our proposed UAVGen on two public UAV object detection datasets. VisDrone [9] is a UAV-view benchmark with 6,471 training, 548 validation, and 1,580 test images, covering 10 categories including pedestrians and vehicles. UAVDT [8] is a UAV detection and tracking benchmark, comprising 24,143 training and 16,592 test images sampled from UAV-captured videos. Both datasets pose core challenges for UAV-based object detection, including abundant

small objects with extreme scale variations, dense targets, frequent occlusions, and dynamic scenes.

We adopt Fréchet Inception Distance (FID) [17] and Average Precision (AP) [13] as our evaluation metrics, where FID measures image generation quality and AP assesses object detection accuracy. For AP, we evaluate mAP, AP₅₀, AP₇₅, AP_s, AP_m and AP_l, following the common size division criteria of MS COCO benchmark [33].

Implementation Details. Our diffusion-driven data synthesis method is built on the FLUX model, trained with a learning rate of 1e-5 for 60K iterations, batch size 8, on one NVIDIA A800 GPU. Training and generation are performed at 512×512 resolution. The diffusion model, text encoder, and VAE encoder are initialized from pretrained FLUX weights and frozen, with remaining parameters finetuned. We adopt Faster R-CNN [37] as the base detector for visual prototype selection and label refinement. For fidelity evaluation, we generate an equal number of images to real data for FID computation; to ensure fair comparison, only VPC-DM is used here, while the full method is adopted for trainability evaluation. On VisDrone, we generate 18,649 patches and merge into 738 full images; on UAVDT, we generate 29,403 patches and merge into 9,802 full images. All compared methods follow standard practice, generating datasets matching the training set size (6,474 for VisDrone, 24,143 for UAVDT). We also reproduce the Copy-Paste baseline [11], with masks inferred via DeepLabV3 ResNet101, synthesizing 7,258 images mixed with real data for GFL detector training on VisDrone.

4.2. Fidelity

We evaluated fidelity of UAVGen on the VisDrone [9] dataset and the UAVDT [8] dataset, by comparing with the latest approaches including GLIGEN [31], Geodiffusion [3]

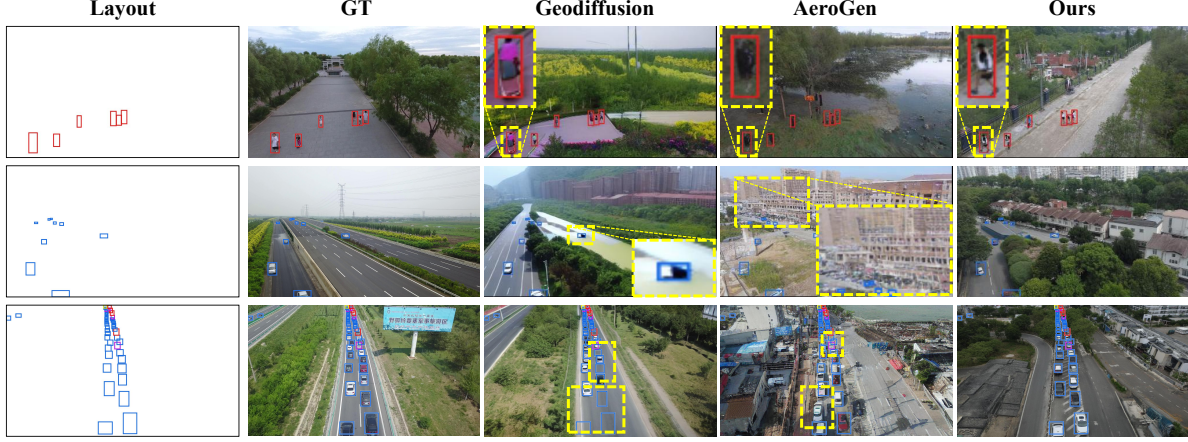


Figure 4. Comparison of generated images on VisDrone. Our method exhibits superior layout-image consistency and enhanced visual fidelity of generated small-scale objects. The yellow dashed boxes denote the regions where the generated targets are inconsistent with the inputs, as well as blurry low-quality targets. For the generation of small pedestrian targets(line 1), our method achieves significantly higher clarity than the comparative methods. Furthermore, in scenarios with dense small targets(line 2, 3), our method exhibits superior consistency between objects and layout.

Table 2. Comparison of Average Precisions (%) achieved by different methods in enhancing the state-of-the-art UAV-based object detection model RemDet-X on VisDrone. The best results are highlighted in **bold**.

Method	mAP $\uparrow$	AP $_{50}\uparrow$	AP $_{75}\uparrow$
Real only	29.8	48.1	30.8
GLIGEN [31]	29.4	47.7	30.2
Geodiffusion [3]	29.4	47.8	30.2
AeroGen [46]	28.8	46.9	29.6
UAVGen (Ours)	30.2	48.7	31.1

and AeroGEN [46]. As represented in Table 1, UAVGen substantially outperforms the compared methods, reducing FID of UAV-based images generation by 13.7 and 2.3 on VisDrone and UAVDT, respectively. This indicates that by leveraging high-quality visual prototypes and fine-grained layout semantics, UAVGen effectively mitigates the challenges posed by small, overlapped objects in UAV imagery, leading to significantly improved generation quality.

We further visualized the synthetic results, as shown in Fig.4. UAVGen demonstrates strong consistency with the input layouts, particularly in challenging scenarios involving small and densely clustered objects. Our focal-region generation strategy effectively improves the generation quality of small objects, producing them with noticeably higher fidelity compared to existing methods.

4.3. Trainability

General Detector Enhancement. For UAV-based object detection, we adopt GFL [30] as the representative general-

purpose model. We trained GFL-ResNet50 for 12 epochs. As shown in Table 1, on VisDrone benchmark, UAVGen achieves state-of-the-art performance with only 738 augmented images, yielding 1.4% improvement in mAP over baseline methods. UAVGen achieves 1.3% improvement in AP $_s$, while the remaining methods show no improvement in small object detection. Furthermore, our approach attains superior mAP across all categories on VisDrone benchmark, with mAP of several categories shown in Fig. 3. On UAVDT benchmark, UAVGen yields 2.1% improvement in mAP and 4.8% improvement in AP $_{50}$. Compared to Geodiffusion [3], our method achieves measurable gains in small-target detection precision, with 1.5% improvement on AP $_s$.

SOTA Detector Enhancement. Beyond conventional detection models, we conducted experiment on RemDet [28], the state-of-the-art detector based on YOLO for the VisDrone dataset. RemDet was trained using default configurations. As shown in Table 2, UAVGen delivers comprehensive performance gains in mAP when applied to RemDet. Notably, comparative methods exhibit detrimental effects on SOTA detector. The experiment demonstrates the efficacy of UAVGen for UAV-based object detection.

4.4. Ablation Study

On VPC-DM. To validate the effectiveness of the data generated by VPC-DM, we conducted ablation experiments on Visual Prototypes and Layout Embeddings. As shown in Table 3, when both components are employed, the detector’s performance improves from 24.5% to 25.2%. Notably, the generated images are detrimental to detector training when neither component is used, while the beneficial impact of the generated data on detector gradually strengthens

Table 3. Ablation study on proposed methods in generation pipeline. "Gen." suggests train detector with generated images, "VP" suggests Visual Prototype, "LE" suggests Layout Embeddings, "FR" suggests Focal Region Generation, "LR" suggests Label Refinement.

VPC-DM			FRE-DP		mAP $\uparrow$	AP $_{50}\uparrow$
Gen.	VP	LE	FR	LR		
					24.5	42.1
✓					23.8	42.1
✓	✓				24.1	42.4
✓	✓	✓			25.2	43.8
✓	✓	✓	✓		25.5	44.5
✓	✓	✓		✓	25.5	44.2
✓	✓	✓	✓	✓	25.9	44.8

Table 4. Ablation study on resolution scale of focal region.

FR Res.	mAP $\uparrow$	AP $_{50}\uparrow$	AP $_{75}\uparrow$
1024	25.3	43.9	25.3
512	25.6	44.7	25.4
256	25.9	44.8	26.0

as the two components are incorporated incrementally.

On FRE-DP. We conducted ablation study on the Focal Region Generation Strategy and Label Refinement, with results summarized in Table 3. Specifically, based on VPC-DM, FRE-DP improves the baseline mAP from 25.2% to 25.9%, indicating the effectiveness of enhancing detection performance. Moreover, both VPC-DM and FRE-DP enhance the quality of the generated images, improving the performance of detector (each brings +0.7%). When both applied concurrently, detector achieves peak mAP 25.9%.

On Synthesized Data Scale. To validate the effectiveness of synthetic images generated by UAVGen, we trained GFL using randomly sampled sets of 100, 300, 500, and 738 (all) synthesized images. As shown in Fig.5, the detector with 100 additional training images achieved performance comparable to AeroGen, which utilizes 6,474 (all) images. The performance of detector exhibits consistent improvement as the number of synthetic training images increases, confirming the effectiveness of our method. Furthermore, our method achieves superior enhancement with fewer synthesized images, demonstrating the efficiency of FRE-DP.

On Region Scale. We also performed an ablation study on resolution within the Focal Region Generation, as summarized in Table 4. We evaluated effect of focal region at different resolutions: 1024, 512, and 256. The results demonstrate a clear trend: as the region resolution decreases, the quality of the generated object regions improves, leading

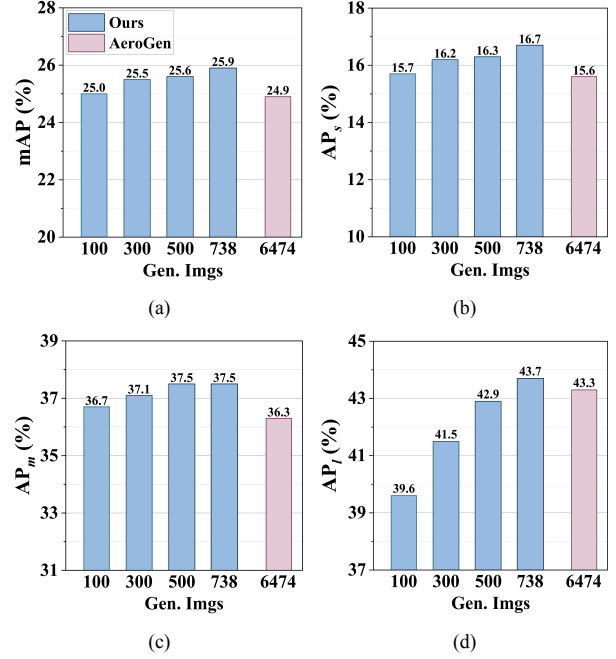


Figure 5. Impact of various scale of generated images on object detection model. With reduced images, our method achieves superior performance.

to enhanced detection performance. When using the smallest resolution of 256, the detector achieves the best results, with mAP 25.9%. This indicates that lower region resolutions can more effectively enhance the quality of small object images and contribute to superior model performance.

5. Conclusion

In this paper, we introduce UAVGen, a diffusion-driven data augmentation framework for UAV-based object detection: we construct high-quality layout conditions for image generation via our Visual Prototype Conditioned Diffusion Model (VPC-DM), focus synthesis on focal regions, and correct misalignments through our Focal Region Enhanced Data Pipeline (FRE-DP), fully exploiting generative potential and enhancing synthesized data consistency. Extensive experiments on diverse datasets demonstrate the effectiveness of our method, which achieves superior image generation fidelity and significantly improves UAV-based detector performance. While tailored for dense small-object detection in UAV scenes, our general-purpose fidelity enhancement and label-image alignment modules extend naturally to general object detection, and FRE-DP's small-object optimization brings notable benefits to far-distant surveillance and remote sensing scenarios, with validation planned for future work. Addressing appearance and scale variations from viewpoint and altitude changes in real-world UAV scenarios remains a promising avenue for further research.

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